02

Lecture 02

2G Circuit Switching

سوئیچ مداری در نسل دوم

2G — 3G — 4G — 5G — 6G

COURSE PART
Part I – 2G/3G foundations (Lectures 1–4)

ESTIMATED STUDY TIME
6–8 hours (3 h reading, 3–5 h exercises)

DIFFICULTY
Intermediate

PREREQUISITES
Lecture 1 (GSM architecture); basic probability (Poisson process, exponential distribution)

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Original lecture notes for the course *Advanced Communication Networks*. The lecture sequence follows the course syllabus (26 lectures, 2G to 6G). Equations use identical notation in the English and Persian editions. Companion edition: Persian (`lecture-02-fa.pdf`).

1 Learning objectives

After studying this lecture you should be able to:

- **LO1** **Explain** what a circuit is, which resources a GSM call reserves and why reservation guarantees quality but wastes capacity.
- **LO2** **Describe** the SS7 protocol stack (MTP, SCCP, TCAP, MAP, ISUP, BSSAP) and the role of signalling points and signal transfer points.
- **LO3** **Trace** a mobile-originated and a mobile-terminated call, including HLR interrogation and the use of the MSRN.
- **LO4** **Derive** the Erlang-B formula from the $M/M/m/m$ model and **dimension** a cell or trunk group for a target blocking probability.
- **LO5** **Quantify** trunking efficiency and the cost of splitting a channel pool.
- **LO6** **Analyse** roaming signalling and call routing, and **evaluate** why circuit switching became a limitation for data.

2 Prerequisite knowledge

You need the GSM architecture and identities of Lecture 1 (MSC, VLR, HLR, GMSC, IMSI, TMSI) and elementary probability: a Poisson arrival process with rate λ, and an exponentially distributed holding time with mean h. We use the word *traffic* in the telephony sense: the average number of simultaneously busy circuits.

Key terms. The vocabulary of this lecture with its Persian equivalents:

Term	Persian	Meaning in this lecture
Circuit switching	سوئیچ مداری	A dedicated path is reserved end to end for the whole call.
Signalling System No. 7 (SS7)	سامانهٔ سیگنالینگ شمارهٔ ۷	Out-of-band signalling network between exchanges.
ISUP	بخش کاربر ISDN	SS7 protocol for setting up and releasing trunk circuits.
MAP	بخش کاربرد موبایل	SS7 application protocol for mobility (HLR, VLR, MSC).
MSISDN	شمارهٔ تلفن مشترک	The dialled telephone number of a subscriber.
MSRN	شمارهٔ رومینگ ایستگاه موبایل	Temporary number used to route a call to the serving MSC.
Erlang (E)	ارلانگ	Unit of traffic: one circuit busy continuously.
Blocking probability	احتمال انسداد	Probability that a call attempt finds all circuits busy.
Grade of service	درجهٔ خدمت	Target blocking probability used for dimensioning.
Trunking efficiency	بازده ترانکینگ	Carried traffic per circuit; grows with pool size.
Roaming	رومینگ	Use of a visited network with a home subscription.

3 Introduction and motivation

Telephony was built on one simple promise: once the called party answers, the voice path behaves like a pair of wires, with constant delay, no loss and no competition from other users. The engineering way to keep that promise is to *reserve* resources for the whole call. GSM inherited this model from ISDN, because its first job was to be a mobile extension of the fixed telephone network and because 1990s exchanges, trunks and billing systems were all built around 64 kbit/s circuits.

Two problems had to be solved on top of the classic telephone network. First, *signalling* had to carry much more than digits: authentication, location updates, handover, roaming between countries. Second, the network had to *find* a subscriber whose location changes, before any circuit can be set up. GSM solved both with the Signalling System No. 7 (SS7) and its Mobile Application Part (MAP). This lecture follows a call from dial to hang-up, then asks the quantitative question every operator faces: *how many circuits are enough?*

4 Fundamental concepts

4.1 What a circuit reserves

For a GSM voice call the network reserves, for its entire duration:

- one traffic channel (TCH) on the radio interface (a timeslot on a carrier);
- one 16 kbit/s sub-channel on the Abis link and one 64 kbit/s timeslot on the A interface after transcoding;
- a connection through the switching matrix of the MSC;
- a 64 kbit/s trunk on each inter-exchange link (one ISUP circuit per link).

The reservation is held even during silence. In a typical conversation each direction is active only about 40% of the time, so most of the reserved capacity carries nothing.

Definition 1 • Traffic intensity

If calls arrive at rate λ (calls per second) and each lasts on average h seconds, the offered traffic is $A = \lambda h$, a dimensionless number measured in erlangs. One erlang is the traffic that keeps one circuit busy all the time.

4.2 In-band versus common-channel signalling

Early telephone networks signalled with tones on the voice circuit itself. This was slow, limited in vocabulary and insecure (tones could be imitated by users). Common-channel signalling separates the two: signalling messages travel over a dedicated packet network, while voice circuits carry only voice. SS7 is the ITU-T standard for this signalling network [6]. It is the first example in this course of the *separation of control plane and user plane*, an idea that will return in every generation.

4.3 Signalling points

Nodes of an SS7 network are *signalling points* (SPs), each identified by a point code. A *service switching point* (SSP) is an exchange such as an MSC. A *signal transfer point* (STP) is a packet router of the SS7 network: it forwards messages between SPs without terminating them, and STPs are deployed in mated pairs for reliability. A *service control point* (SCP) hosts databases and service logic, for example the intelligent-network logic used for prepaid charging.

5 Detailed technical explanation

5.1 The SS7 protocol stack

Figure 1 shows the stack as used in GSM.

- **MTP 1–3** (Message Transfer Part) carries messages reliably between adjacent nodes (MTP 2: framing, error detection and retransmission) and routes them across the signalling network using point codes (MTP 3).
- **SCCP** (Signalling Connection Control Part) adds connection-less and connection-oriented services and, crucially for mobility, *global title translation*: a message can be addressed with

a telephone-number-like global title (for example derived from an IMSI) and the network finds the point code of the HLR.

- **TCAP** (Transaction Capabilities Application Part) structures request–response dialogues between applications.
- **MAP** [3] runs over TCAP and defines mobility operations: Update Location, Insert Subscriber Data, Send Authentication Info, Send Routing Information (SRI), Provide Roaming Number (PRN), Cancel Location.
- **ISUP** (ISDN User Part) runs directly over MTP 3 and sets up and releases trunk circuits between exchanges with messages such as IAM, ACM, ANM, REL and RLC.
- **BSSAP** runs over SCCP on the A interface. Its BSSMAP part carries messages between MSC and BSC (assignment, handover, ciphering); its DTAP part transparently carries messages between MSC and MS (call control and mobility management).

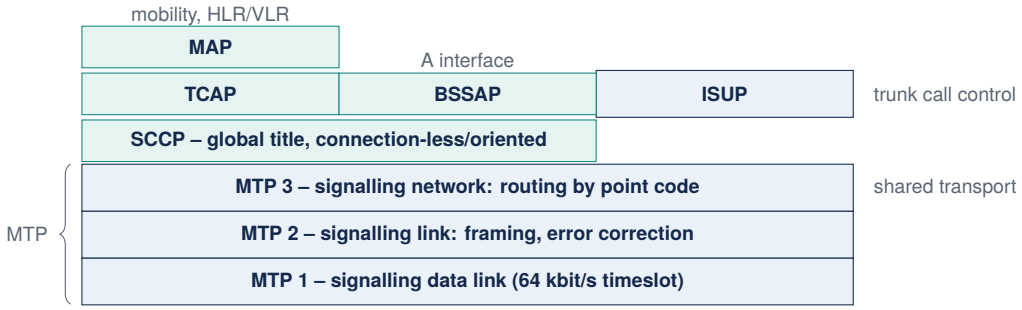


Figure 1. SS7 stack in a GSM core network. MTP provides transport; ISUP controls trunk circuits; SCCP and TCAP support the MAP mobility dialogues and BSSAP on the A interface.

5.2 Layer-3 protocols between MS and MSC

On the radio side, layer 3 is divided into three sublayers [4]: *radio resource management* (RR, terminated in the BSS: channel assignment, handover), *mobility management* (MM: location updating, authentication, TMSI reallocation) and *connection management* (CM, with call control CC: Setup, Alerting, Connect, Disconnect). MM and CC messages are carried transparently through the BSS in DTAP.

5.3 Numbers and identities used for routing

Three numbers must not be confused:

- **MSISDN** – the number people dial; it points to the subscriber's home network and is used by the GMSC to find the HLR.
- **IMSI** – the subscription identity used inside the mobile network (MAP addressing, authentication); never dialled.
- **MSRN** – a temporary E.164 number allocated by the visited VLR for one incoming call; it routes the call through the fixed network to the serving MSC.

The HLR maps MSISDN to IMSI and stores which VLR serves the subscriber; the VLR maps the MSRN back to the IMSI when the call arrives.

5.4 Location updating

Three cases trigger a location update [2]:

- **Normal** – the MS reads a new location area identity (LAI) on the broadcast channel. If the new LA belongs to another VLR, the new VLR fetches the subscriber data from the HLR with Update Location; the HLR sends Cancel Location to the old VLR.
- **Periodic** – the MS re-registers when timer T3212 expires, so the network can purge subscribers that disappeared without detaching (battery removed, coverage lost).
- **IMSI attach** – at power-on in the same LA. The mirror procedure, **IMSI detach**, tells the VLR that the MS is off, so incoming calls are not paged in vain.

5.5 Roaming

A roaming subscriber registers in a visited network (VPLMN). The visited VLR addresses the home HLR (HPLMN) over the international SS7 network, using global titles derived from the IMSI, and receives the subscription data and authentication triplets. Roaming agreements define charging and which services are allowed. Two consequences are worth noting:

- **Authentication stays with the home network**: only the home AuC knows K_i.
- **Tromboning**: a call to a roamer is always routed first to the home GMSC (the MSISDN belongs to the home network) and then internationally to the visited MSC. A local caller in the visited country therefore pays for two international legs, unless optimal routing is implemented.

5.6 Call release

Either party can release. The mobile sends Disconnect; the MSC replies Release and the MS confirms Release Complete. Towards the fixed network the MSC sends ISUP REL and receives RLC, which frees the trunk. Finally the MSC orders the BSS to release the radio channel (BSSMAP Clear Command). Billing records are closed when the circuit is released, which is why circuit-switched charging is naturally time-based.

6 Mathematical models

6.1 The loss system and the Erlang-B formula

Engineering problem. A cell has m traffic channels. Calls that arrive when all are busy are *blocked* (lost); they do not wait. How large must m be so that no more than, say, 2% of attempts are blocked?

System model and assumptions. (i) Call arrivals form a Poisson process of rate λ from a large population; (ii) holding times are independent and exponentially distributed with mean $h = 1/\mu$; (iii) blocked calls are cleared (no retry); (iv) all m channels are fully accessible. This is the $M/M/m/m$ queue.

Derivation. Let p_k be the steady-state probability that k channels are busy. In state k the system moves up at rate λ and down at rate $k\mu$. Balancing the flow between neighbouring states gives

$$\lambda\, p_{k-1} = k\mu\, p_k \quad\Longrightarrow\quad p_k = \frac{A}{k}\, p_{k-1} = \frac{A^k}{k!}\, p_0, \qquad A = \frac{\lambda}{\mu} = \lambda h. \tag{1}$$

Normalising $\sum_{k=0}^{m} p_k = 1$ yields $p_0 = \left(\sum_{i=0}^{m} A^i/i!\right)^{-1}$. Because Poisson arrivals see time averages (the PASTA property), the probability that an arriving call is blocked equals the probability that all m channels are busy:

$$P_{\mathrm{B}} = B(A,m) = \frac{A^m/m!}{\displaystyle\sum_{i=0}^{m} \frac{A^i}{i!}}. \tag{2}$$

Symbols and units. A: offered traffic [E]; m: number of channels; λ: arrival rate [calls/s]; h: mean holding time [s]; P_{B}: blocking probability (dimensionless). The carried traffic is $A_{\mathrm{c}} = A(1-P_{\mathrm{B}})$ and the utilisation of each channel is $\rho = A_{\mathrm{c}}/m$.

Numerically stable form. Direct evaluation of Eq. (2) overflows for large m. Dividing numerator and denominator by $A^m/m!$ gives the recursion

$$B(A,0) = 1, \qquad B(A,k) = \frac{A\,B(A,k-1)}{k + A\,B(A,k-1)}, \quad k = 1,\dots,m, \tag{3}$$

used in `scripts/acnmath.py` to produce Figure 2 and every number below.

Interpretation. Equation (2) depends only on the *mean* holding time (it is insensitive to the holding-time distribution), which is why it worked well for telephony for a century. It assumes, however, that blocked users go away; in reality many retry, which increases the offered traffic and the true blocking (Exercise 5).

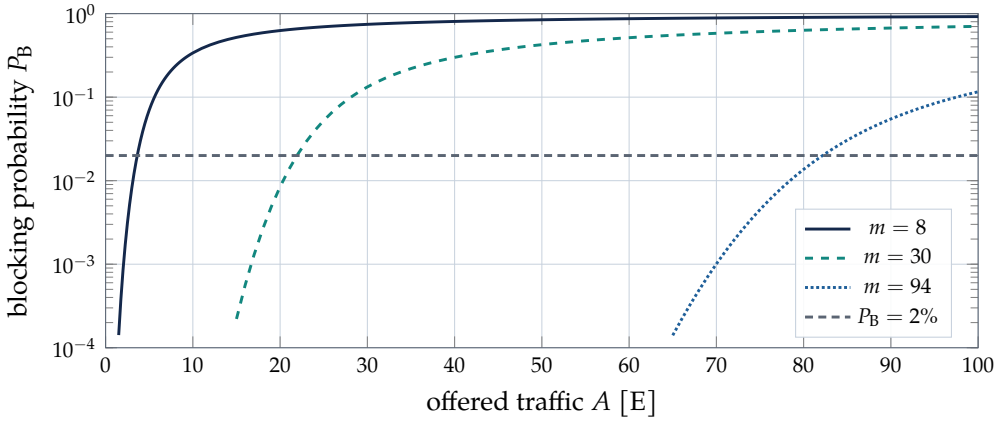


Figure 2. Erlang-B blocking probability versus offered traffic for $m = 8$, 30 and 94 channels. For the same 2% grade of service, 8 channels carry 3.6 E (0.45 E per channel) whereas 94 channels carry 82.2 E (0.87 E per channel).

6.2 Trunking efficiency

Problem. Is it better to have one pool of $2m$ channels or two pools of m?

Result. For a fixed grade of service, large pools carry more traffic per channel. This is the *trunking gain* (or pooling gain) and follows from Eq. (2): the relative fluctuation of the number of active calls, $\sqrt{A}/A$, shrinks as A grows, so less spare capacity is needed. Table 1 quantifies it. It explains why GSM operators avoid splitting channels between sectors unless interference requires it, and why statistical multiplexing in packet networks is so attractive.

Table 1. Traffic carried at 2% blocking (computed with Eq. (3)).

Channels m	Offered traffic A [E]	Carried traffic per channel [E]	Utilisation
8	3.63	0.44	44%
15	9.01	0.59	59%
30	21.93	0.72	72%
47	37.46	0.78	78%
94	82.17	0.86	86%

6.3 Waste of a reserved circuit

If the voice activity factor is α (fraction of time a speaker is active), the useful load of a reserved circuit is α times its capacity. With $\alpha \approx 0.4$ in each direction, roughly 60% of every reserved circuit-second carries silence. Packet switching recovers this capacity statistically, at the price of variable delay – the fundamental trade-off between the two switching paradigms, revisited in Lectures 3 and 5.

7 Network architecture and diagrams

The SS7 network is a separate packet network overlaid on the circuit network. Every MSC, GMSC and HLR is a signalling point; STPs route between them. Trunks between MSCs and the PSTN carry voice, while MAP and ISUP messages travel on signalling links that may follow completely different physical routes. Figure 4 shows the interplay for the hardest case, an incoming call, and Figure 3 the outgoing case.

8 Step-by-step procedures

Procedure 1 • Mobile-originated call (MOC)

1. The MS sends a channel request on the random access channel (RACH) and receives a signalling channel (SDCCH) on the access grant channel (AGCH).
2. It sends CM Service Request; the MSC/VLR authenticates the MS and starts ciphering (Lecture 1).
3. The MS sends Setup with the called number. The MSC checks the subscription (barring, prepaid credit) and answers Call Proceeding.
4. The MSC orders the BSS to assign a traffic channel (BSSMAP Assignment Request); the

MS moves from SDCCH to a TCH.

5. The MSC analyses the number and sends ISUP IAM towards the destination; the remote exchange answers ACM when the called phone rings, and the MSC sends Alerting to the MS (ring-back tone).
6. When the called party answers, ISUP ANM arrives; the MSC sends Connect, the MS acknowledges, and charging starts.

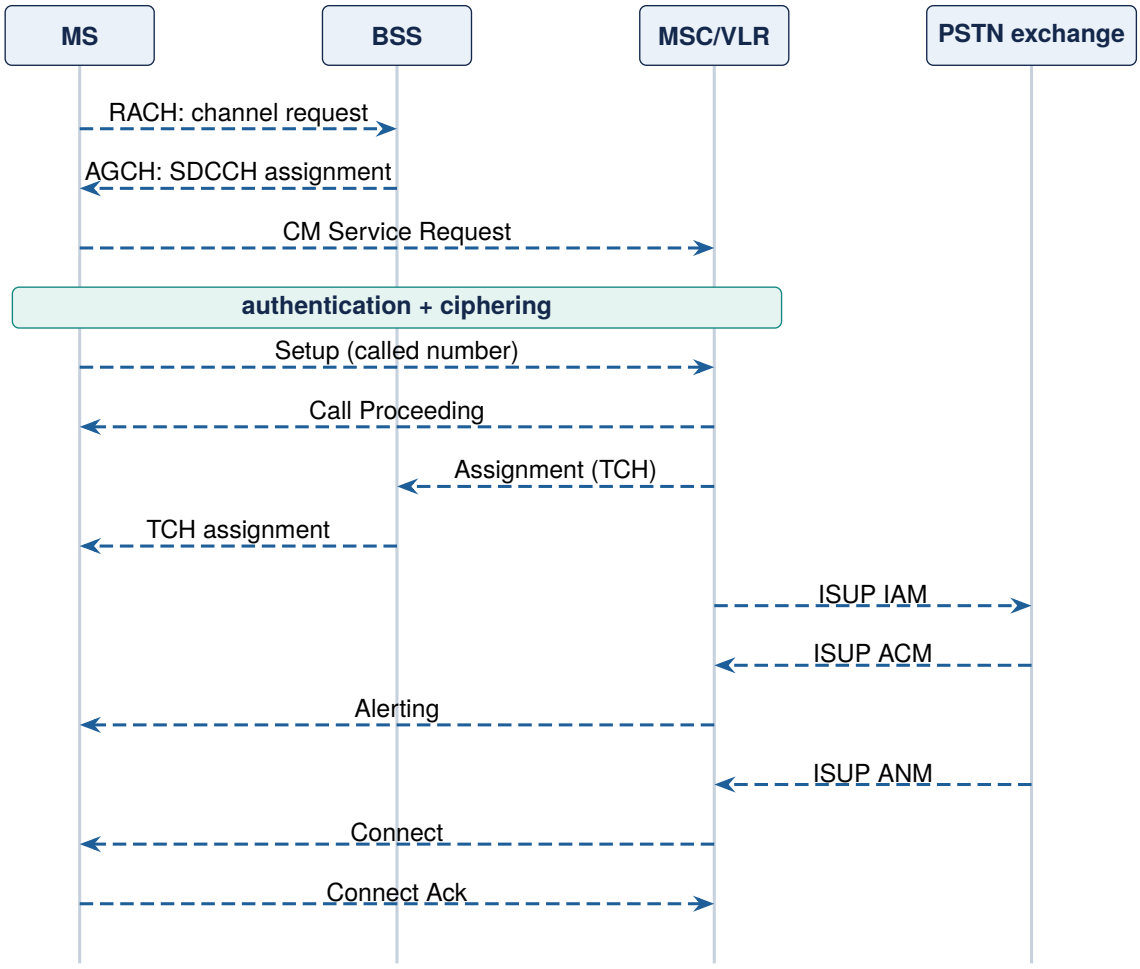


Figure 3. Mobile-originated call (Procedure 1).

Procedure 2 • Mobile-terminated call (MTC)

1. A caller in the PSTN dials the MSISDN. The PSTN routes the call (ISUP IAM) to a GMSC of the home network.
2. The GMSC sends MAP Send Routing Information to the HLR.
3. The HLR finds the serving VLR and requests a roaming number with MAP Provide Roaming Number; the VLR allocates an MSRN and returns it.
4. The HLR returns the MSRN to the GMSC, which routes the call with a new IAM addressed to the MSRN – the fixed network delivers it to the serving MSC.
5. The MSC asks its VLR for the subscriber context; the VLR releases the MSRN and returns the location area; the MSC pages the MS in all cells of that LA.

6 The MS answers the page, is authenticated, receives Setup and a TCH; ringing and answer are signalled back with ACM and ANM.

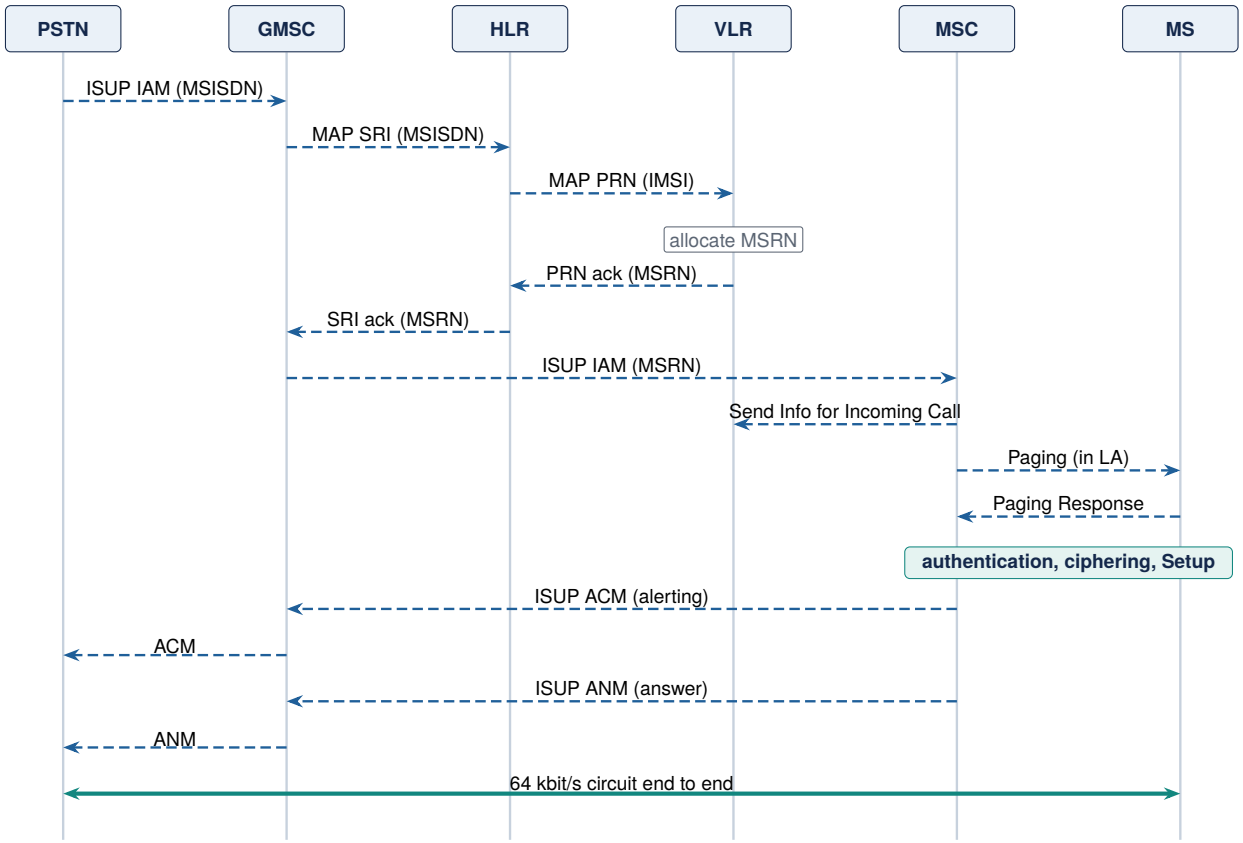


Figure 4. Mobile-terminated call: HLR interrogation, MSRN allocation, routing and paging (Procedure 2). Dashed: signalling; solid: the reserved circuit.

9 Worked examples

Worked example 1 • Dimensioning a cell

Problem. The cell of Lecture 1, Example 1, has 94 traffic channels. Subscribers generate 25 mE each during the busy hour. How many subscribers can the cell serve at a 2% grade of service?
Solution. Solving $B(A, 94) = 0.02$ with Eq. (3) gives $A = 82.2$ E. The number of subscribers is $N = A/a = 82.2/0.025 \approx 3290$. Carried traffic $A_c = 82.2 \times 0.98 = 80.5$ E, i.e. 86% channel utilisation.
Interpretation. If the same 94 channels were split into two sectors of 47, each sector would carry 37.5 E, 74.9 E in total – about 9% fewer subscribers for the same hardware (Table 1).

Worked example 2 • Erlang-B by hand

Problem. A small site has $m = 3$ channels and $A = 2$ E. Compute P_B.

Solution.

$$P_B = \frac{2^3/3!}{1 + 2 + 2^2/2! + 2^3/3!} = \frac{1.333}{6.333} = 0.211.$$

The recursion gives the same: $B_1 = 2/3$, $B_2 = (2 \cdot \frac{2}{3})/(2 + \frac{4}{3}) = 0.4$, $B_3 = 0.8/(3 + 0.8) = 0.211$.

Interpretation. One call in five is lost: a 3-channel site is useless at 2 E. Small pools need very low load for an acceptable grade of service.

Worked example 3 • Busy-hour arrival rate

Problem. Mean call duration is 90 s. What arrival rate produces the 82.2 E of Example 1, and how many call set-ups per second must the MSC handle in that cell?

Solution. $\lambda = A/h = 82.2/90 = 0.913$ calls/s, i.e. about 3290 call attempts per hour in one cell. Each attempt generates tens of signalling messages (RR, MM, CC, BSSMAP and ISUP), so signalling load scales with λ, not with A.

Interpretation. Short calls (or SMS and location updates, which use SDCCH) stress signalling capacity, which must be dimensioned separately (Exercise 4).

10 Technology comparison

Table 2. Circuit switching versus packet switching.

	Circuit switching (GSM voice)	Packet switching (GPRS and later)
Resource allocation	Reserved at set-up for the whole call	Per packet or per scheduling interval
Delay	Constant, low	Variable (queuing)
Loss mechanism	Blocking at set-up	Packet loss or delay under load
Efficiency	Low for bursty sources	High (statistical multiplexing)
Dimensioning model	Erlang-B ($M/M/m/m$)	Queueing models ($M/M/1$, etc.)
Charging	Time	Volume or flat rate
Signalling	SS7: ISUP + MAP	GTP, later Diameter and HTTP/2

Why circuit switching was right for voice. Voice is a constant-rate, delay-sensitive stream; reserving 13 kbit/s guarantees quality without complex scheduling, and the Erlang model allows precise dimensioning.

Why it became a limitation. Data is bursty: a web page needs a high rate for a second and nothing for a minute. Reserving a circuit for that is wasteful, and a single 9.6 kbit/s circuit is far too slow. Running two core networks (circuit and packet) also doubled operational cost – the motivation for the all-IP core of LTE (Lecture 5), where voice itself becomes packets (VoLTE).

11 Key takeaways

Key takeaways

- A GSM call reserves a radio timeslot, Abis/A channels, a switch path and trunks for its whole duration – even during silence.
- SS7 separates signalling from voice: MTP transports, SCCP addresses (global titles), TCAP structures dialogues, MAP handles mobility, ISUP controls trunks.
- An incoming call needs the HLR (SRI) and the visited VLR (PRN) to obtain an MSRN before it can be routed and paged.
- Erlang-B, $B(A, m)$, dimensions loss systems; compute it with the recursion Eq. (3).
- Larger pools are more efficient (trunking gain); splitting pools costs capacity.
- Circuit switching wastes the silent part of every call and cannot serve bursty data efficiently; this drove the move to packet switching.

12 Exercises

Exercise 1 • Level 1 – conceptual • Signalling

Why does ISUP run directly over MTP 3, while MAP needs SCCP? Relate your answer to how the destination of each message is known.

Exercise 2 • Level 1 – conceptual • Numbers

Why can the GMSC not route an incoming call using the IMSI? Why is the MSRN released as soon as the call reaches the serving MSC?

Exercise 3 • Level 2 – numerical • Dimensioning

A cell has 30 traffic channels. (a) What traffic can it offer at 1% and at 5% blocking? (b) With 20 mE per subscriber, how many more subscribers are served at 5% than at 1%?

Exercise 4 • Level 2 – numerical • SDCCH capacity

Location updates, SMS and call set-ups each hold an SDCCH for about 3 s. A cell sees 1.5 such transactions per second in the busy hour. How many SDCCH sub-channels are needed for a blocking of at most 0.5%?

Exercise 5 • Level 3 – analytical • Retries

Assume every blocked caller retries once, immediately and independently. Show that the offered traffic becomes approximately $A' = A(1 + P'_{\mathrm{B}})$ where $P'_{\mathrm{B}} = B(A', m)$, solve this fixed-point equation numerically for $m = 30$, $A = 21.93$ E, and comment on the result.

Exercise 6 • Level 3 – analytical • Tromboning

A subscriber of an Iranian operator roams in Turkey and is called by a Turkish fixed-line user. Draw the circuit path and list which networks carry each leg. Propose a signalling change that would avoid the international legs, and identify what information the Turkish network would need.

Solutions

Solution to Exercise 1

ISUP messages travel between adjacent exchanges along the circuit being set up; the next exchange and its point code are known from the routing tables, so MTP 3 routing is enough. MAP messages are addressed to databases identified by subscriber identities (e.g. "the HLR of IMSI 432..."), possibly in another country; SCCP global title translation converts such an identity into a point code, hop by hop.

Solution to Exercise 2

The PSTN routes by E.164 numbers; an IMSI is not a dialable number and is deliberately not exposed outside the mobile network. The MSRN is a scarce temporary number taken from the visited MSC's pool; once the call reaches that MSC the IMSI is retrieved from the VLR and the MSRN can be reused for another call.

Solution to Exercise 3

(a) $B(A,30) = 0.01 \Rightarrow A = 20.3\,\mathrm{E}$; $B(A,30) = 0.05 \Rightarrow A = 24.8\,\mathrm{E}$. (b) Subscribers: $20.3/0.02 \approx 1017$ versus $24.8/0.02 \approx 1240$, i.e. about 220 more (22%) at the price of five times more blocking.

Solution to Exercise 4

$A = \lambda h = 1.5 \times 3 = 4.5\,\mathrm{E}$. Iterating Eq. (3): $B(4.5,10) = 0.0105$, $B(4.5,11) = 0.0043$. Eleven sub-channels are needed; in practice operators configure two SDCCH/8 channels (16 sub-channels), which also leaves room for peaks.

Solution to Exercise 5

A fraction P'_{B} of the λ first attempts generate a second attempt, so the total attempt rate is $\lambda(1 + P'_{\mathrm{B}})$ and $A' = A(1 + P'_{\mathrm{B}})$ (treating retries as Poisson, which is an approximation). Iterating from $P'_{\mathrm{B}} = 0.02$: $A' = 22.37$, $B = 0.0236$; $A' = 22.45$, $B = 0.0243$; converging to $A' \approx 22.47\,\mathrm{E}$, $P'_{\mathrm{B}} \approx 0.0244$. The blocking seen by an attempt grows by about 22%; the fraction of *users* who finally fail is $P'^{2}_{\mathrm{B}} \approx 0.06\%$, but the network carries more signalling. Retries are a positive feedback that can destabilise an overloaded network.

Solution to Exercise 6

Path: Turkish caller → Turkish PSTN → international leg to the home GMSC in Iran (the MSISDN is Iranian) → HLR interrogation returns a Turkish MSRN → second international leg back to the visited MSC in Turkey → the roamer. Optimal routing lets the Turkish gateway query the home HLR directly (MAP SRI towards the HPLMN) and route to the MSRN inside Turkey; it requires the visited network to recognise the number as belonging to a roaming

subscriber and an agreement allowing the HLR query, and it changes charging, which is why it was rarely deployed.

13 Further reading

SS7 is introduced in [6]; MAP is specified in [3], location management in [2] and call handling in [1]. Layer-3 call control and mobility messages are in [4]. Kleinrock [7] derives the loss formula rigorously, and Rappaport [8] applies it to cellular dimensioning. Eberspächer et al. [5] describe GSM call flows in detail.

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